

Exercise #1. (50 points)

Consider the Gen 3 Kinova 6DOF robot shown here. We do not follow the “DH Table” from the User’s Guide (uploaded), as it does not follow the DH convention and has mistakes in it. From the schematic diagram, and keeping the sign and directions of the axes as in the User’s Guide, we can derive the following DH Table, where $l_0 = 0.1564 + 0.1284$, $l_2 = 0.410$, $l_3 = 0.0064 - 0.0054$, $l_4 = 0.2083 + 0.1059$, $l_6 = 0.1059 + 0.0615$.

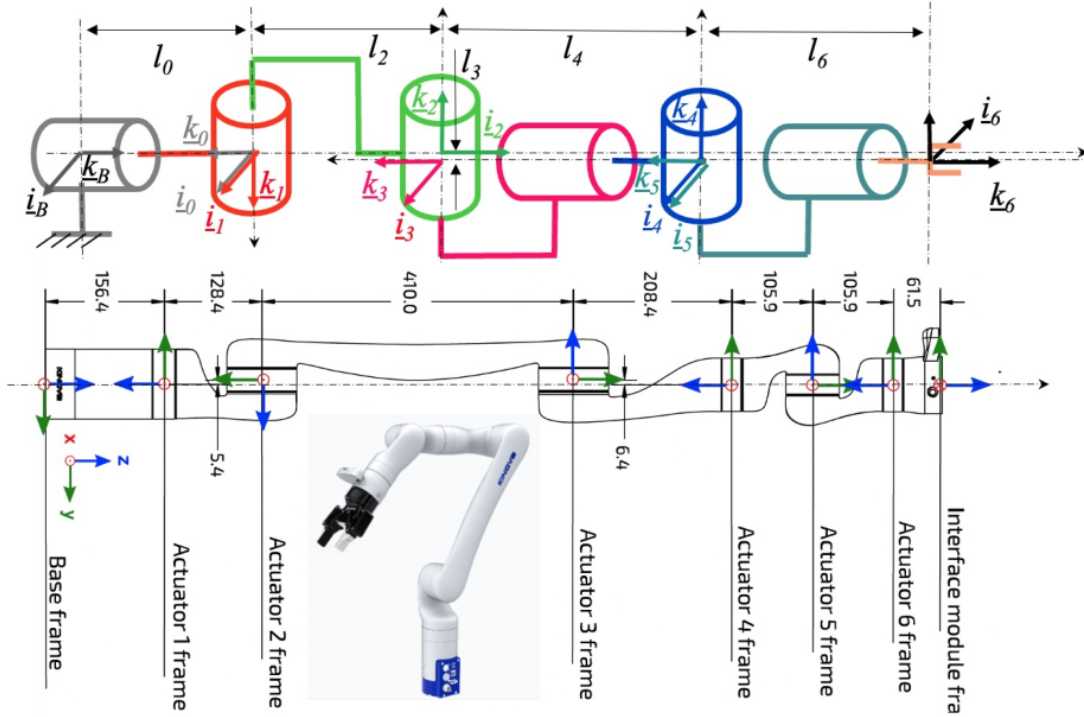


Table 1: Kinova Gen3 Classical DH parameters with joint limits

i	θ_i (rad)	d_i (m)	a_i (m)	α_i (rad)	Joint limits (deg)
0 (from base)	0.0	l_0	0.0	π	—
1	(q_1)	0.0	0.0	$\pi/2$	$(-\infty, +\infty)$
2	$(q_2) - \pi/2$	0.0	l_2	π	$[-128.9^\circ, 128.9^\circ]$
3	$(q_3) - \pi/2$	$-l_3$	0.0	$-\pi/2$	$[-147.8^\circ, 147.8^\circ]$
4	(q_4)	$-l_4$	0.0	$-\pi/2$	$(-\infty, +\infty)$
5	(q_5)	0.0	0.0	$\pi/2$	$[-120.3^\circ, 120.3^\circ]$
6 (to interface)	$(q_6) + \pi$	$-l_6$	0.0	π	$(-\infty, +\infty)$

(a) Write a MATLAB function to find the forward kinematics homogeneous matrix ${}^B T_6(q)$ from the

DH table, giving coords of $\{\tilde{o}_6, C_6\}$ in $\{\tilde{o}_B, C_B\}$.

(b) Find the manipulator geometric Jacobian. In which robot configuration is the robot Jacobian singular?

Solution.

Every joint is spherical, so we have a manipulator jacobian of the form:

$$J = \begin{bmatrix} k_0 \times (\tilde{o}_6 - \tilde{o}_0) & k_1 \times (\tilde{o}_6 - \tilde{o}_1) & k_2 \times (\tilde{o}_6 - \tilde{o}_2) & k_3 \times (\tilde{o}_6 - \tilde{o}_3) & k_4 \times (\tilde{o}_6 - \tilde{o}_4) & k_5 \times (\tilde{o}_6 - \tilde{o}_5) \\ k_0 & k_1 & k_2 & k_3 & k_4 & k_5 \end{bmatrix}.$$

Since joints 4, 5, and 6 form a spherical wrist, the origins $\tilde{o}_3$, $\tilde{o}_4$, and $\tilde{o}_5$ may be taken to coincide for the purpose of the Jacobian calculation. In particular, we use $\tilde{o}_3 = \tilde{o}_4 = \tilde{o}_5$. Also, from the geometry of the manipulator, the axes k_1 and k_2 are antiparallel, so that $k_2 = -k_1$, and the first two frame origins coincide: $\tilde{o}_0 = \tilde{o}_1$.

Using these observations, the geometric Jacobian can be written in the equivalent form

$$J \sim_{\text{eq}} \begin{bmatrix} k_0 \times (\tilde{o}_6 - \tilde{o}_0) & k_1 \times (\tilde{o}_6 - \tilde{o}_0) & -k_1 \times (\tilde{o}_6 - \tilde{o}_2) & k_3 \times (\tilde{o}_6 - \tilde{o}_3) & k_4 \times (\tilde{o}_6 - \tilde{o}_3) & k_5 \times (\tilde{o}_6 - \tilde{o}_3) \\ k_0 & k_1 & -k_1 & k_3 & k_4 & k_5 \end{bmatrix}.$$

Now add to the first block row the second block row multiplied by $(\tilde{o}_6 - \tilde{o}_3)$. Then, J is singular if and only if the following matrix is singular:

$$\hat{J} \sim_{\text{sing}} \begin{bmatrix} k_0 \times (\tilde{o}_3 - \tilde{o}_0) & k_1 \times (\tilde{o}_3 - \tilde{o}_0) & -k_1 \times (\tilde{o}_3 - \tilde{o}_2) & 0 & 0 & 0 \\ k_0 & k_1 & -k_1 & k_3 & k_4 & k_5 \end{bmatrix}.$$

This is singular iff at least two of the column vectors are linearly dependent. We analyze the robot to see when this can be the case. First, though k_4 will always be perpendicular to k_3 and k_5 , in the case of wrist gimbal lock we can have $k_3 \parallel k_5$.

Now, since $\tilde{o}_3$ is offset from k_0 , it can never be the case that $\tilde{o}_3 - \tilde{o}_2 \parallel \tilde{o}_3 - \tilde{o}_0$, $k_1 \parallel (\tilde{o}_6 - \tilde{o}_0)$ or $k_1 \parallel (\tilde{o}_6 - \tilde{o}_2)$, or $k_0 \parallel (\tilde{o}_3 - \tilde{o}_0)$, so we have no singularities arising from the first 3 joints.

Therefore, the only singularity happens when $k_3 \parallel k_5$.

(c) Write a MATLAB function to compute the robot Jacobian for a given set of joint angles. For example, in response to a function call or to user input, e.g.,

$$q = [45^\circ \quad -45^\circ \quad 45^\circ \quad 0^\circ \quad -30^\circ \quad 90^\circ],$$

the program will output the manipulator Jacobian, as a 6×6 matrix of coordinates relative to the base coordinate system $\{\tilde{o}_B, C_B\}$, relating joint velocities to end-effector velocities in base frame.

Use your code to compute the manipulator Jacobian and its condition number for a number of interesting sets of joint configurations, including

$$q = [0^\circ \quad 90^\circ \quad 90^\circ \quad 0^\circ \quad 90^\circ \quad 0^\circ]^T,$$

$$q = [0^\circ \quad 90^\circ \quad 90^\circ \quad 0^\circ \quad 0^\circ \quad 0^\circ]^T,$$

$$q = [0^\circ \quad 7.2767^\circ \quad 15^\circ \quad 0^\circ \quad 90^\circ \quad 0^\circ]^T,$$

and comment on the results.

Note: It may be useful to render the Kinova using the Robotics Toolbox by Peter Corke.

Exercise #2. (50 points)

Write four Matlab functions that solve the respective Kahan problems. The returned angles should be in degrees. Include the proper number of solutions or return NaN (not a number) for an invalid solution. Apply a reasonable tolerance threshold when determining if a solution exists.

$$\theta = \text{KahanP1}(s, t)$$

finds the angle between s and t ($\theta \in [0, 180^\circ]$).

$$\theta = \text{KahanP2}(\hat{s}, u, v)$$

solves

$$e^{\theta \hat{s} \times} u = v$$

for $\theta \in (-180^\circ, 180^\circ]$.

$$[\phi, \theta] = \text{KahanP3}(\hat{s}, \hat{t}, u, v)$$

solves

$$e^{\theta \hat{s} \times} u = e^{\phi \hat{t} \times} v$$

with $(\theta, \phi \in (-180^\circ, 180^\circ])$.

$$\theta = \text{KahanP4}(a, b, c)$$

finds the supplementary angle formed by sides a and b of triangle abc ($\theta \in [0, 180^\circ]$).

Exercise #3: (50 points)

(a) Solve the inverse kinematics of the Kinova 6DOF manipulator.

(b) Write a MATLAB program that provides all inverse kinematics solutions. Inputs are the coordinates in base frame of the desired end-effector location ($\tilde{o}_d$), approach vector (k_d) and sliding vector (j_d).

Output all valid sets of joint angles (in degrees) that achieve this.

Verify that your program works for several distinct sets of inputs, including the joint variables listed under Problem 1(c), by feeding each of your solutions into the forward kinematics (briefly explain how you chose the inputs).

Document your code with references to your inverse kinematics solution. Carefully consider how many possible solutions there are, considering the joint ranges in the table provided.

Hint: Since $\tilde{o}_4$ is the centre of the wrist, we have

$$\begin{aligned}\tilde{o}_4 - \tilde{o}_0 &= e^{-\theta_1 k_B \times} e^{\theta_2 j_B \times} \left[l_2 k_B + e^{-\theta_3 j_B \times} (l_4 k_B + l_3 j_B) \right] \\ &= e^{-\theta_1 k_B \times} e^{\theta_2 j_B \times} \left[l_2 k_B + l_3 j_B + e^{-\theta_3 j_B \times} l_4 k_B \right]\end{aligned}$$

where we moved $l_3 j_B$ outside the rotation $e^{-\theta_3 j_B \times}$ since $l_3 j_B$ is aligned with the rotation axis.

Projecting on the plane $k_B - i_B$ we have Kahan's problem P4. Solve for θ_3 , then we are left with Kahan's problem P3.

Solution. (a) We follow the hint, and start with:

$$\tilde{o}_4 - \tilde{o}_0 = e^{-\theta_1 k_B \times} e^{\theta_2 j_B \times} \left[l_2 k_B + e^{-\theta_3 j_B \times} (l_4 k_B + l_3 j_B) \right].$$

Premultiplying by $e^{-\theta_2 j_B \times} e^{\theta_1 k_B \times}$, we have

$$e^{-\theta_2 j_B \times} e^{\theta_1 k_B \times} (\tilde{o}_4 - \tilde{o}_0) = l_2 k_B + e^{-\theta_3 j_B \times} (l_4 k_B + l_3 j_B). \quad (1)$$

However, j_B doesn't rotate about j_B , so we have

$$e^{-\theta_2 j_B \times} e^{\theta_1 k_B \times} (\tilde{o}_4 - \tilde{o}_0) = l_2 k_B + l_3 j_B + e^{-\theta_3 j_B \times} l_4 k_B.$$

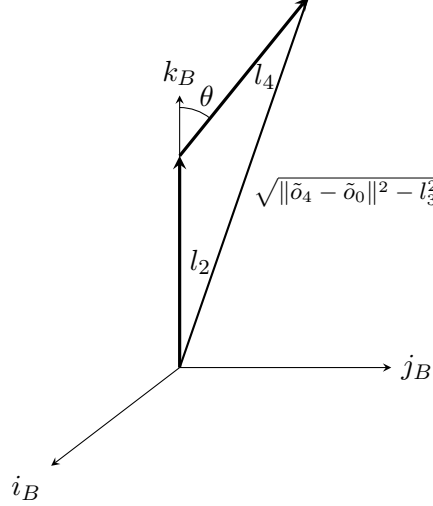
Next, we subtract $l_3 j_B$ from both sides.

$$e^{-\theta_2 j_B \times} e^{\theta_1 k_B \times} (\tilde{o}_4 - \tilde{o}_0) - l_3 j_B = l_2 k_B + e^{-\theta_3 j_B \times} l_4 k_B.$$

The vector on the right is purely in the k_B - i_B plane, so we can apply pythagorus to solve the length of the quantity on the left:

$$\|e^{-\theta_2 j_B \times} e^{\theta_1 k_B \times} (\tilde{o}_4 - \tilde{o}_0) - l_3 j_B\|^2 = \|\tilde{o}_4 - \tilde{o}_0\|^2 - l_3^2.$$

Thus, we have a triangle with sidelengths l_2 , l_4 , and $\sqrt{\|\tilde{o}_4 - \tilde{o}_0\|^2 - l_3^2}$, with the triangle then has an exterior angle θ_3 outside the sidelengths l_2 and l_4 .



This is a KP4 problem, which we then solve for θ_3 . For reference in the computer program, I will write this out explicitly:

$$|\theta_3| = KP_4(l_2, l_4, \sqrt{\|\tilde{o}_4 - \tilde{o}_0\|^2 - l_3^2}) \quad (2)$$

With θ_3 , we can then compute two possible solutions $w_{\pm} = l_2 k_B + l_3 j_B + e^{\mp \theta_3 j_B \times} (l_4 k_B + l_3 j_B)$. Then, equation (1) gives us

$$\begin{aligned} e^{-\theta_2 j_B \times} e^{\theta_1 k_B \times} (\tilde{o}_4 - \tilde{o}_0) &= w_{\pm} \\ e^{\theta_1 k_B \times} (\tilde{o}_4 - \tilde{o}_0) &= e^{\theta_2 j_B \times} w_{\pm}. \end{aligned}$$

This is a KP3 problem. For each vector $w_{\pm}$ we can solve for potentially 2 solutions that are pairs (θ_1, θ_2) .

The KP_3 problem is stated here:

$$u = \tilde{o}_4 - \tilde{o}_0 \quad (3)$$

$$w_{\pm} = l_2 k_B + l_3 j_B + e^{\mp \theta_3 j_B \times} l_4 k_B \quad (4)$$

$$(\theta_1, \theta_2)_i = KP_3(u, w_{\pm}, k_B, j_B). \quad (5)$$

Since the range of θ_2 is limited, these solutions must be tested to ensure that they obey the constraints.

Now, having solved for potential values $\theta_1, \theta_2, \theta_3$, we fix a potential solution, and then use forward kinematics to compute C_3, o_3 in the frame C_B, o_B . Following the notes, we have that C_6 can be obtained from C_3 following a series of rotations:

$$\underline{C}_6 = \underline{C}_3 e^{\theta_4 k \times} e^{\theta_5 j \times} e^{\theta_6 k \times}.$$

Next, we re-arrange, and consider the vector j in each co-ordinate frame (which we are given, since this is the second column of $\underline{C}_6$ and $\underline{C}_3$) respectively:

$$\begin{aligned}\underline{C}_6 e^{-\theta_6 k \times} &= \underline{C}_3 e^{\theta_4 k \times} e^{\theta_5 j \times} \\ e^{-\theta_6 k \times} \underline{C}_6 &= e^{\theta_4 k \times} \underline{C}_3 e^{\theta_5 j \times} \\ e^{-\theta_6 k \times} \underline{C}_6 j &= e^{\theta_4 k \times} \underline{C}_3 e^{\theta_5 j \times} j \\ e^{-\theta_6 k \times} \underline{C}_6 j &= e^{\theta_4 k \times} \underline{C}_3 j \\ e^{-\theta_6 k \times} \underline{j}_6 &= e^{\theta_4 k \times} \underline{j}_3.\end{aligned}$$

This is a KP_3 problem, which we can solve for θ_4 , and θ_6 candidates. That is:

$$\theta_4, -I\theta_6 = KP_3(k_6, j_6, k_3, j_3).$$

Knowing θ_4 , we use forward kinematics to solve for $\underline{C}_4 = \underline{C}_3 e^{\theta_4 k \times}$, and

$$\begin{aligned}\underline{C}_6 e^{-\theta_6 k \times} &= \underline{C}_4 e^{\theta_5 j \times} \\ \underline{C}_6 e^{-\theta_6 k \times} k &= \underline{C}_4 e^{\theta_5 j \times} k \\ k_6 &= e^{\theta_5 j \times} k_4.\end{aligned}$$

Since we have C_4 , we know j_4 , k_4 , and since we have C_6 , we know k_6 . This is a KP_2 problem, so we have

$$\theta_5 = KP_2(\underline{j}_4, \underline{k}_4, \underline{k}_6). \tag{6}$$

With this process, we can then compute a list of possible candidates for $\vec{\theta} = (\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$.